Title: Pedestrian Detection in Urban Environments – Implications for Autonomous Vehicle Safety
We Came, We Saw, We Modelled
Names: Emily Friedman (group leader), Jensen Harvey, and Kaitlyn Chou
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Goal Statement: We will build an object detection model that identifies and localizes pedestrians in urban street images with at least 85% detection accuracy.

Research Question: How accurately can image-based models detect pedestrians in urban street scenes, and what are the implications of detection errors for autonomous vehicle safety?

Model Approach: We will develop an object detection model using a Faster R-CNN architecture with a ResNet18 backbone to identify and localize pedestrians in urban street images. The model will be trained and fine-tuned on a subset of the Cityscapes dataset, leveraging existing annotations to detect pedestrians within each image. Faster R-CNN is selected because it is a widely used and effective approach for object detection tasks, enabling both classification and localization of objects, which is more representative of real-world autonomous vehicle perception systems than simple image classification. The ResNet18 backbone will be used for feature extraction, allowing the model to efficiently learn relevant visual patterns while reducing computational complexity through transfer learning. Model performance will be evaluated using metrics such as precision, recall, and detection accuracy, with particular emphasis on recall to assess the model's ability to avoid missed pedestrian detections (false negatives). This approach is appropriate because object detection models better capture the spatial and safety-critical nature of pedestrian recognition in autonomous driving contexts.

Executive Summary: This project develops an object detection model to identify and localize pedestrians in urban street scenes using the Cityscapes dataset. The analysis focuses on evaluating model performance with particular attention to safety-critical errors such as missed pedestrian detections. The results will provide insight into the reliability of image-based perception systems in autonomous vehicle contexts.

Data Set Summary: The dataset we will use in this project is the Cityscapes dataset, specifically the `gtBboxCityPersons` annotations, which provides bounding box labels for pedestrian detection. The dataset consists of 3,475 annotated images, with 2,975 images in the training set and 500 in the validation set. These images were collected from 21 cities and represent diverse urban environments with varying levels of pedestrian density and scene complexity.

Exploratory analysis shows that approximately 84.6% of images contain at least one pedestrian, while 15.4% contain none, indicating a moderate class imbalance. Additionally, the dataset includes over 25,000 pedestrian annotations and approximately 8,400 ignore-region annotations, highlighting the complexity of real-world urban scenes. Bounding box analysis

reveals that pedestrian sizes and visibility vary significantly, with many small or partially occluded instances, which presents a meaningful challenge for object detection models.

Data Dictionary:

- **file**: File path corresponding to the image and annotation
- **split**: Dataset partition indicating whether the image belongs to the training or validation set
- **city**: City where the image was collected (e.g., Frankfurt, Munich)
- **img_width / img_height**: Dimensions of the image in pixels
- **label**: Object category for each annotation (e.g., pedestrian, rider, ignore region)
- **bbox_x / bbox_y**: Top-left coordinates of the bounding box
- **bbox_w / bbox_h**: Width and height of the bounding box in pixels
- **bbox_vis_w / bbox_vis_h**: Width and height of the visible (non-occluded) portion of the bounding box
- **is_person**: Indicator variable identifying valid pedestrian-related annotations
- **is_ignore**: Indicator variable for ignore regions in the dataset
- **has_pedestrian**: Image-level binary variable indicating whether at least one pedestrian is present
- **n_persons**: Number of pedestrian annotations in a given image

Questions Explored in EDA:

How are images distributed across cities and dataset splits?

- We produced the “Image Count by City and Split” plot to understand geographic coverage and identify whether certain cities are overrepresented.
- This helps assess dataset diversity and potential bias across locations.

What is the overall structure of the dataset?

- Using the “Number of Images per Dataset Split” plot, we analyzed how images are distributed between training and validation sets.
- This ensures there is sufficient data for both model training and evaluation.

What is the distribution of annotation labels?

- From the “Annotation Label Counts Across All Images” plot, we explored the frequency of different object labels (like pedestrian, rider, ignore region).
- This helps identify which classes dominate the dataset and confirms that pedestrian annotations are the primary focus.

What proportion of images contain at least one pedestrian?

- The “Image-Level Pedestrian Presence” plot was used to determine how often pedestrians appear in images.
- This establishes a binary perspective of the data and reveals class imbalance between images with and without pedestrians.

How does pedestrian presence vary across cities?

- Using the “Pedestrian Presence Rate by City” plot, we analyzed variation in pedestrian frequency across different cities.
- This helps identify geographic differences in scene composition that may affect model performance.

How many pedestrians appear per image?

- The “Distribution of Person Count per Image” histogram was used to examine how pedestrian counts vary across images.
- This provides insight into scene complexity and highlights the presence of both sparse and crowded environments.

What are the characteristics of pedestrian bounding boxes?

- From the bounding box distribution plots (width, height, area, and aspect ratio), we analyzed variation in pedestrian size and shape.
- This helps identify challenges such as small or irregularly shaped objects that may be difficult for the model to detect.

Current Unknowns or Questions:

How well will the model detect small or distant pedestrians?

- EDA shows wide variation in bounding box size, with many small objects.
- It is unclear whether the model will reliably detect pedestrians that occupy only a small portion of the image.

How does occlusion affect detection performance?

- Visibility analysis indicates that many pedestrians are partially occluded.
- It remains uncertain how significantly occlusion will reduce detection accuracy and recall.

How will class imbalance impact model performance?

- A large majority of images contain pedestrians (~84.6%)
- This imbalance may bias the model or affect its ability to generalize.

What detection confidence threshold should be used?

- Object detection models require a confidence threshold to determine valid predictions.
- The optimal threshold balancing precision and recall has not yet been determined.

How will performance vary across different cities or environments?

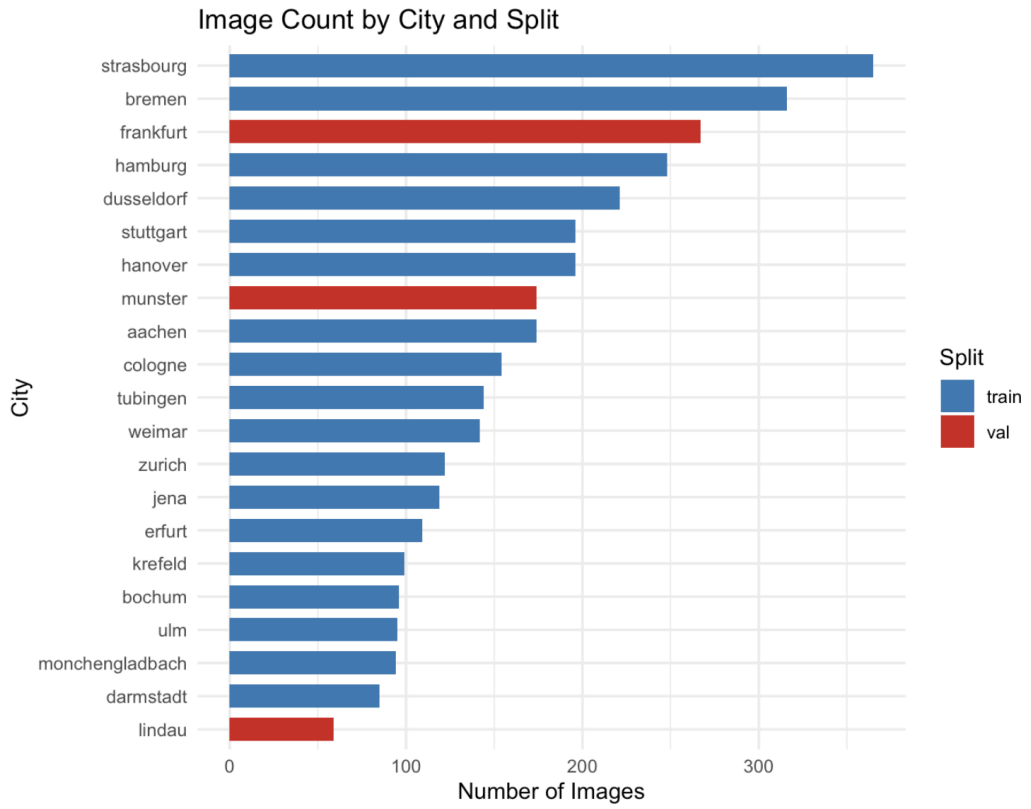
- EDA shows variation in pedestrian presence across cities.
- It is unclear whether the model will perform consistently across different urban environments.

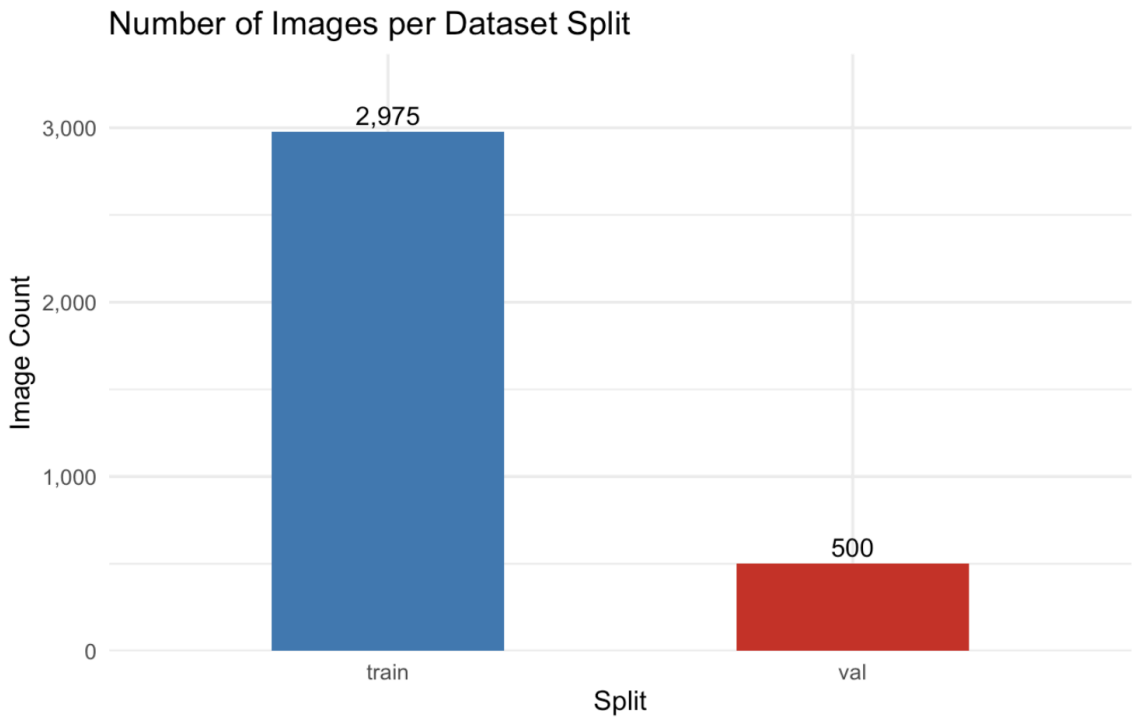
How should evaluation prioritize safety-critical errors?

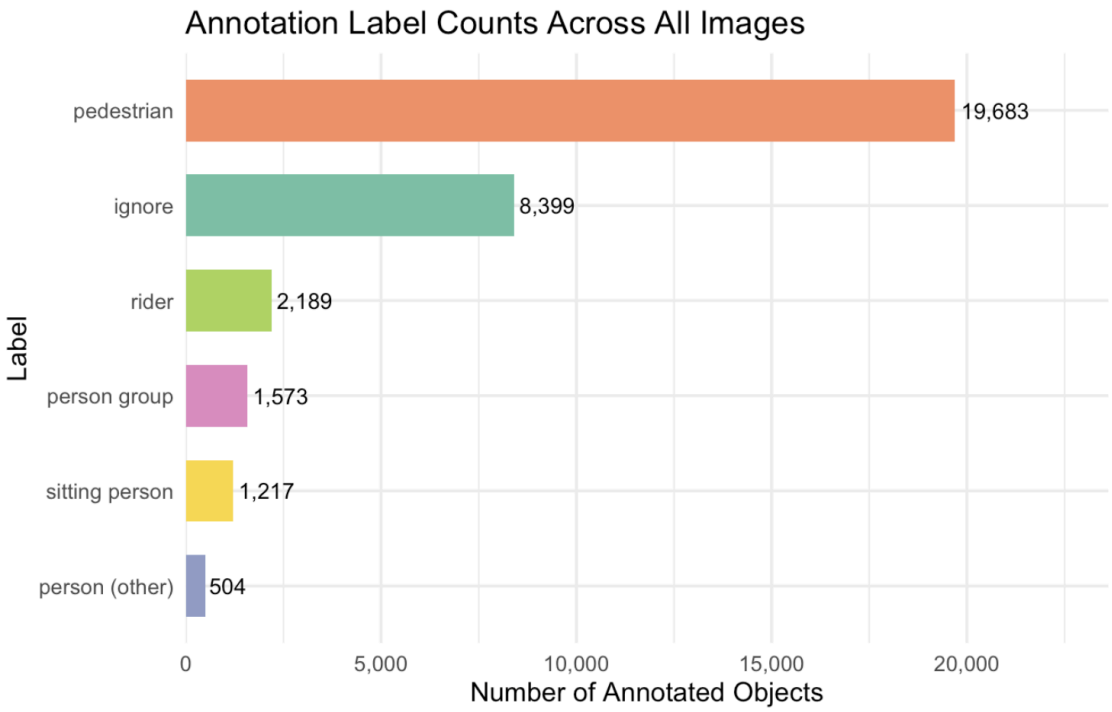
- EDA and project motivation highlight the importance of false negatives (missed pedestrians).
- We will have to decide how to balance overall accuracy with recall to reflect real-world safety concerns.

Refinement of Research Question/Model Plan: EDA did not necessitate a change to the core research question or model architecture, but it did sharpen several implementation decisions. The 5.51:1 class imbalance confirmed that a standard cross-entropy loss is insufficient; we will use a weighted loss function or focal loss to prevent the model from over-predicting the majority (pedestrian-present) class during background/negative image classification. Additionally, the wide range of bounding box sizes (median area 2,520 px^2, max exceeding 100,000 px^2) informed our decision to include multi-scale anchor boxes in the RPN rather than relying on a fixed anchor configuration. Finally, the confirmation that all images share a uniform 2048×1024 resolution simplifies preprocessing, as no resizing normalization is needed across the dataset.

Exploratory Plots:






Annotation Label Counts Across All Images
pedestrian
ignore
rider
person group
sitting person
person (other)
19,683
8,399
2,189
1,573
1,217
504
Label
0
5,000
10,000
15,000
20,000
Number of Annotated Objects

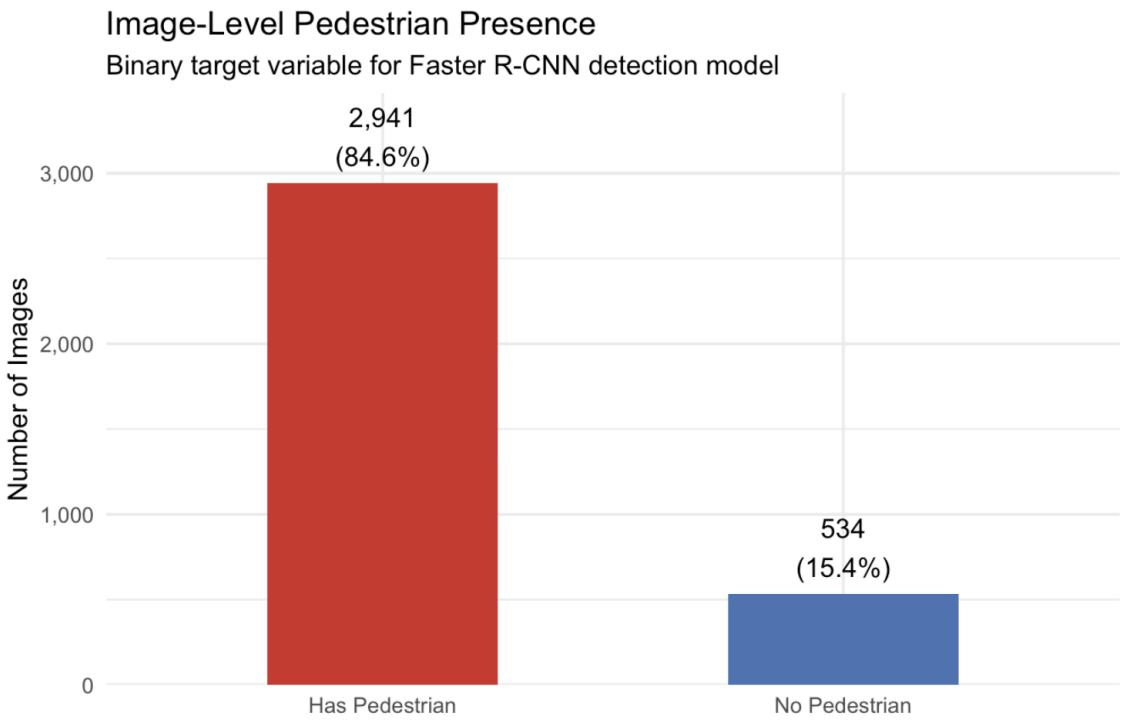
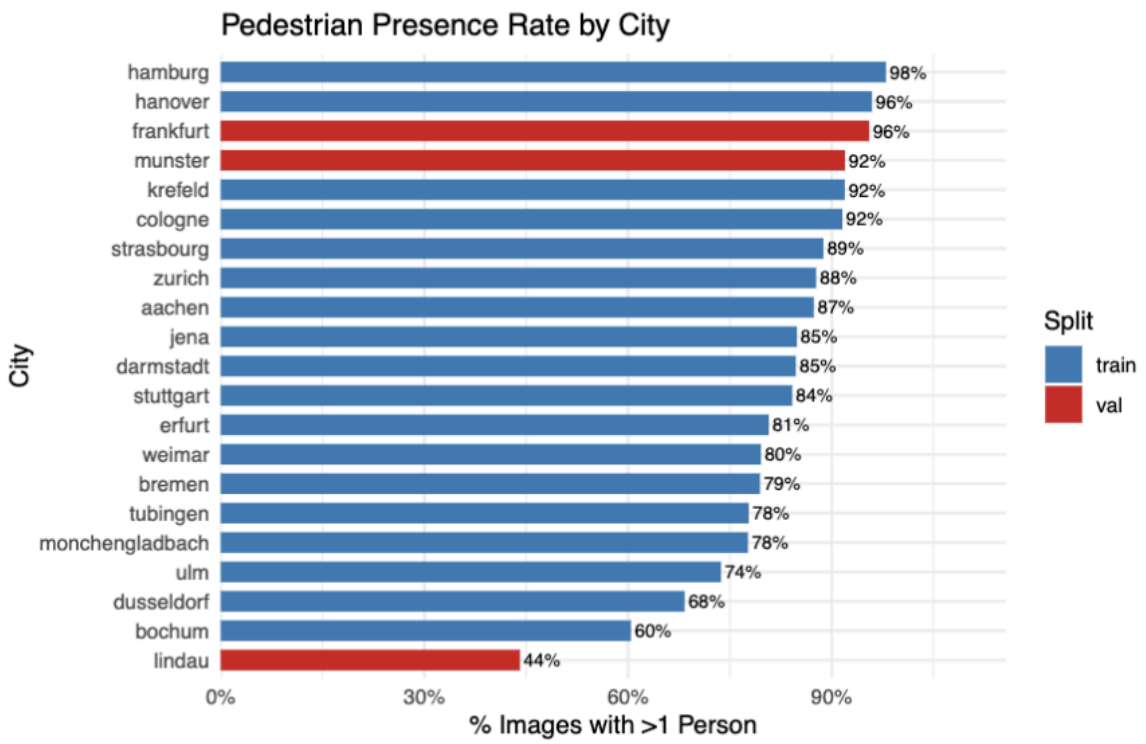
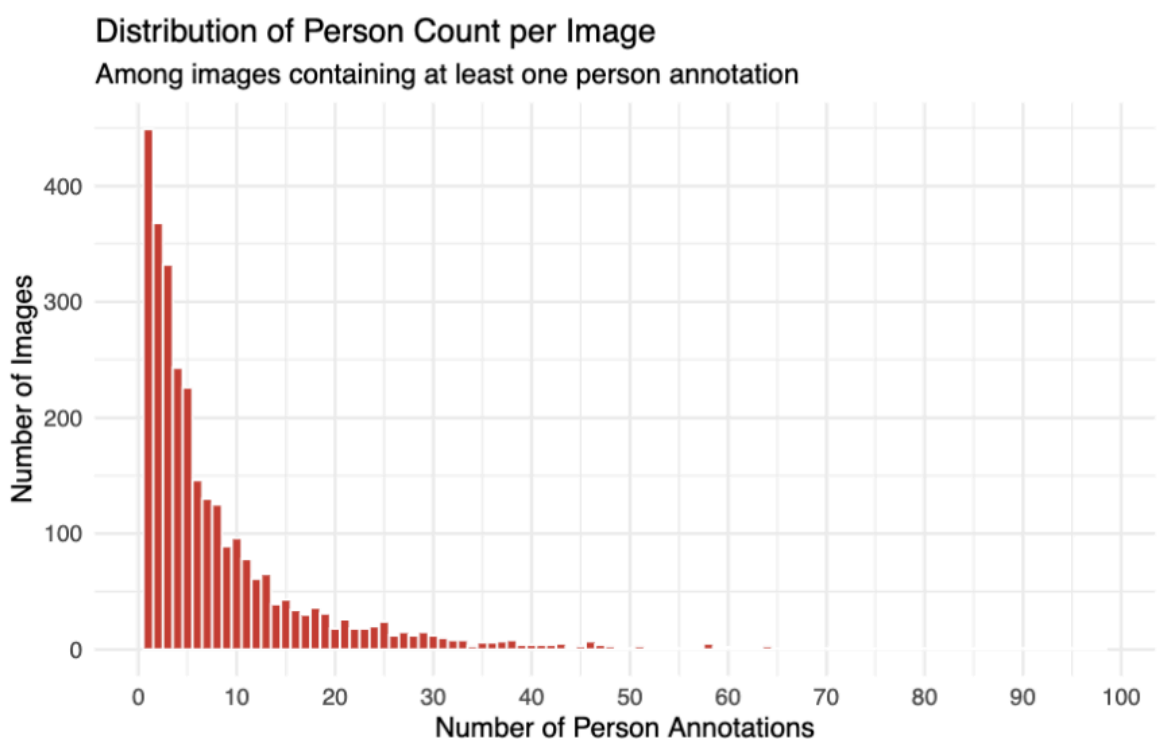
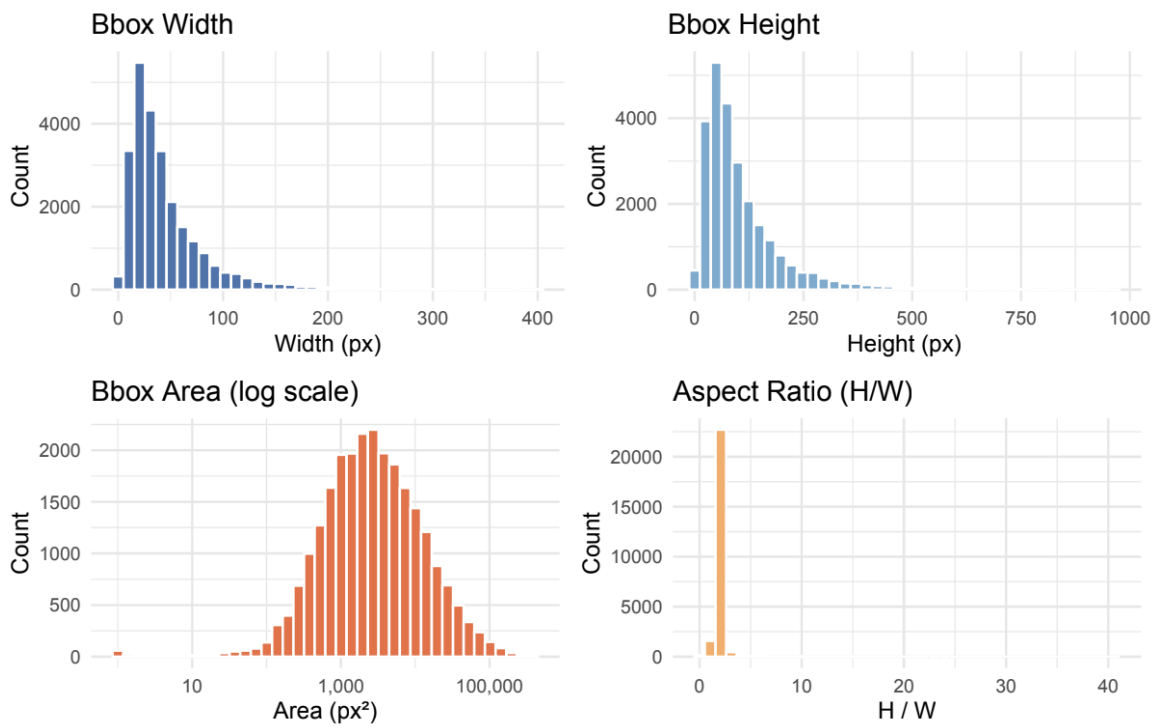


Image-Level Pedestrian Presence
Binary target variable for Faster R-CNN detection model
2,941
(84.6%)
534
(15.4%)
Number of Images
3,000
2,000
1,000
0
Has Pedestrian
No Pedestrian

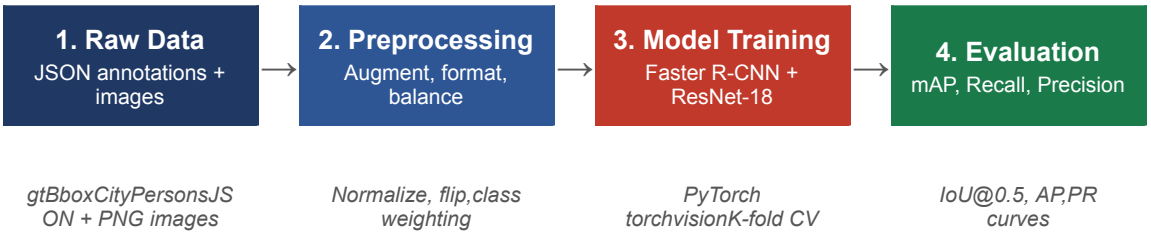

Pedestrian Presence Rate by City
hamburg 98%
hanover 96%
frankfurt 96%
munster 92%
krefeld 92%
cologne 92%
strasbourg 89%
zurich 88%
aachen 87%
jena 85%
darmstadt 85%
stuttgart 84%
erfurt 81%
weimar 80%
bremen 79%
tubingen 78%
monchengladbach 78%
ulm 74%
dusseldorf 68%
bochum 60%
lindau 44%
City
0%
30%
60%
90%
% Images with >1 Person
Split
train
val


Distribution of Person Count per Image
Among images containing at least one person annotation
Number of Images
0
100
200
300
400
0
10
20
30
40
50
60
70
80
90
100
Number of Person Annotations



Analysis Plan:
Graphic:

Analysis Pipeline



Preprocessing: The Cityscapes PNG images and gtBboxCityPersons JSON annotations will be loaded and converted into PyTorch tensors using torchvision transforms. All images will be normalized using ImageNet channel means and standard deviations (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]) given the pretrained ResNet-18 backbone. Data augmentation will be applied to the training set to improve generalization: random horizontal flips (p=0.5), random brightness and contrast jitter (±20%), and small-scale random cropping. Ignore-region annotations (label="Ignore") will be masked from the loss computation during training so that ambiguous regions do not penalize the model. To address the 5.51:1 class imbalance identified in EDA, a weighted sampling strategy will be used during training batch construction, ensuring negative images (no pedestrian) are not systematically underrepresented. Finally, bounding box annotations will be converted to the [x_min, y_min, x_max, y_max] format required by torchvision's detection API.

Methodology: The model will follow a region-based object detection pipeline using PyTorch and torchvision. A Faster R-CNN model will be initialized with a pretrained ResNet-18 backbone (ImageNet weights), and the Region Proposal Network (RPN) will generate candidate bounding boxes from multi-scale feature maps. The RPN anchor configuration will include small anchor scales (16, 32, 64, 128 px) to account for the many small pedestrians identified in EDA. The detection head will then classify each proposal and refine bounding box coordinates. The full model will be fine-tuned end-to-end on the Cityscapes training split (2,975 images) using the Adam optimizer with an initial learning rate of 1e-4 and step-based learning rate decay. K-fold cross-validation (k=5) will be applied within the training set to tune hyperparameters including learning rate, batch size, and anchor scales. The existing Cityscapes validation split (500 images) will serve as the held-out test set for final evaluation and will not be used during hyperparameter tuning.

Evaluation: Model performance will be evaluated using the following metrics computed on the held-out validation split. The primary evaluation metric is recall (sensitivity) for the pedestrian class, which directly measures the model's ability to avoid missed detections (the safety-critical error in autonomous vehicle contexts). Secondary metrics include mean Average Precision (mAP) at IoU threshold 0.5 (mAP@0.5) and overall precision. Precision-recall curves will be generated at varying confidence thresholds to identify the operating point that best balances safety and precision. Per-city recall will also be reported to assess whether performance varies systematically across different urban environments. Bounding boxes will be considered true positives when the intersection-over-union (IoU) with a ground-truth annotation exceeds 0.5.

Quantifiable Goal: The goal of this project is to develop a pedestrian detection model using a ResNet-18 backbone within an R-CNN framework that achieves 85% detection accuracy for the pedestrian class on a held-out test set.

Goal Put in Another Way: The goal of this project is to develop a model that determines whether a pedestrian is in front of you.

References:

- [1] M. Cordts, M. Omran, S. Ramos, T. Rehfeld, M. Enzweiler, R. Benenson, U. Franke, S. Roth, and B. Schiele, "The Cityscapes Dataset for Semantic Urban Scene Understanding," in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), Las Vegas, NV, Jun. 2016.
- [2] S. Zhang, R. Benenson, and B. Schiele, "CityPersons: A Diverse Dataset for Pedestrian Detection," in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), Honolulu, HI, Jul. 2017.